PROBLEM 15 | Problems T2 defects.pdf | Crystal Defects & Solid Solutions

Vacancy Fraction in Aluminium near the Melting Point

Original statement · Problems T2 defects.pdf

1. The fraction of vacancies in an aluminum lattice is 2.29×10^{-5} *at* $400^{\circ}\mathrm{C}$*. Calculate the fraction of vacancies at* $660^{\circ}\mathrm{C}$ *(just before the melting point).* $R = 8.31\ \mathrm{J/(mol \cdot K)}$*. (Solution:* 4.53×10^{-4}*)*

Solution: 4.53×10^{-4}

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Material: Pure aluminium (Al, FCC lattice).
- Temperature 1: $T_1 = 400^{\circ}\mathrm{C} = 400 + 273.15 = 673.15\ \mathrm{K}$.
- Vacancy fraction at T_1:

$$\left(\frac{n_v}{N}\right)_1 = 2.29 \times 10^{-5}$$

- Temperature 2: $T_2 = 660^{\circ}\mathrm{C} = 660 + 273.15 = 933.15\ \mathrm{K}$ (temperature just below the melting of aluminium, $T_m \approx 660.3^{\circ}\mathrm{C}$).
- Universal gas constant: $R = 8.31\ \mathrm{J/(mol \cdot K)}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

The thermodynamic equilibrium concentration of vacancies in a metallic lattice follows the Boltzmann distribution [Session 4 Slide 16]:

$$\frac{n_v}{N} = \exp\left(-\frac{\Delta H_v}{RT}\right)$$

where ΔH_v is the enthalpy of formation of one mole of vacancies in the lattice.

Solution Strategy:

1. With the data at T_1, determine the vacancy formation enthalpy ΔH_v:

$$\ln\left(\frac{n_v}{N}\right)_1 = -\frac{\Delta H_v}{R \cdot T_1} \implies \Delta H_v = -R \cdot T_1 \cdot \ln\left(\frac{n_v}{N}\right)_1$$

2. With ΔH_v constant (independent of temperature in this range), compute the new fraction at T_2:

$$\left(\frac{n_v}{N}\right)_2 = \exp\left(-\frac{\Delta H_v}{R \cdot T_2}\right)$$

Or, combining into a single two-point relation:

$$\ln\left(\frac{(n_v/N)_2}{(n_v/N)_1}\right) = -\frac{\Delta H_v}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

`PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP`

1. Formation Enthalpy (ΔH_v):

- Natural logarithm of the initial fraction:

$$\ln(2.29 \times 10^{-5}) = -10.68437$$

- Solving for ΔH_v:

$$\Delta H_v = -(8.31\ \text{J/mol} \cdot \text{K}) \times (673.15\ \text{K}) \times (-10.68437)$$

$$\Delta H_v = 5593.8765 \times 10.68437 = 59767.1\ \text{J/mol} \approx \mathbf{59.77}\ \text{kJ/mol}$$

(In atomic units: $E_v = \frac{59767.1}{6.022\times10^{23}\times1.602\times10^{-19}} \approx 0.62$ eV/atom, consistent with the literature for aluminium).

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2. Vacancy Fraction at 660°C ($T_2 = 933.15\ \text{K}$):

- Boltzmann exponent at T_2:

$$\frac{\Delta H_v}{R \cdot T_2} = \frac{59767.1\ \text{J/mol}}{(8.31\ \text{J/mol}\cdot\text{K}) \times (933.15\ \text{K})} = \frac{59767.1}{7754.4765} = 7.70743$$

- Equilibrium vacancy fraction:

$$\left(\frac{n_v}{N}\right)_2 = \exp(-7.70743) = \mathbf{4.49 \times 10^{-4}}$$

Check with $T = 273 + t$ (i.e. $T_1 = 673\ \text{K}, T_2 = 933\ \text{K}$): $\Delta H_v = 8.31 \times 673 \times 10.68437 = 59754\ \text{J/mol}$ and $(n_v/N)_2 = \exp(-59754/(8.31 \times 933)) = \exp(-7.7069) = 4.50 \times 10^{-4}$.

Discrepancy with the official solution. The official key gives 4.53×10^{-4}. Recomputing from the given data (2.29×10^{-5} at 400°C, $R = 8.31\ \text{J/(mol K)}$) gives $\exp(-7.7074) = 4.49 \times 10^{-4}$ with $T = 273.15 + t$, and $\exp(-7.7069) = 4.50 \times 10^{-4}$ with $T = 273 + t$ ($\Delta H_v = 59.75\ \text{kJ/mol}$). The key's 4.53×10^{-4} ($\approx 0.8\%$ higher) is not reproduced by either convention; no value was adjusted.

`PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION`

- **Exponential Thermal Increase:** Going from 400°C to 660°C ($\Delta T = 260^\circ\text{C}$), the vacancy concentration is multiplied by a factor of:

$$\frac{4.49 \times 10^{-4}}{2.29 \times 10^{-5}} \approx \mathbf{19.6}\ \text{times}$$

- **Physical Limit of Real Crystals:** The fraction reached near melting ($4.49 \times 10^{-4} \approx$ 1 vacancy per 2200 atoms) satisfies the general rule quoted in the official slide [Session 4 Slide 16]:

$$\frac{n_v}{N} \sim 10^{-4}\ \text{maximum}$$

This high density of thermal vacancies near the melting point is responsible for the drastic increase in the rate of atomic diffusion in homogenisation and aeronautical sintering processes.